Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/30/2014
PI Name:	Sowa, Marianne Ph.D.		
Project Title:	Integrated experimental and computational study of radiation induced matrix remodeling in a human skin equivalent		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	marianne.sowa@pnl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	509-371-6898
Organization Name:	Battelle Pacific Northwest National Laboratory		
PI Address 1:	Biological Sciences Division		
PI Address 2:	PO BOX 999, MS J4-02		
PI Web Page:			
City:	Richland	**State:**	WA
Zip Code:	99352-0999	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	11/01/2009	**End Date:**	04/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 4/30/2014 per NSSC information (Ed., 4/5/2013) NOTE: End date changed to 4/30/2013, per NSSC information (Ed., 8/30/2012) NOTE: Period of performance changed to 11/1/09-10/31/12 (from 10/9/09-10/8/12) per NSSC info (3/2010) NOTE: Period of performance changed to 10/9/09-10/8/12 (from 11/15/09-11/14/12) per NSSC info (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shankaran, Harish (Battelle, Pacific Northwest Divsion)		
Grant/Contract No.:	NNX10AB06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Current risk estimates for manned space flight are based predominantly on human epidemiology data following low linear energy (LET) radiation exposures. These data are scaled to the radiation types and fluences commonly encountered in space through the application of radiation quality factors and dose, dose rate modification factors. NASA and the scientific community acknowledge the limitations of this approach, however appropriate data on the radiation quality dependence for high energy, high atomic number ions (HZE) such as those found in space are currently unavailable making for large uncertainties in risk estimates. This application proposes an integrated experimental and computational strategy to develop a predictive understanding of the link between cell signaling, gene expression and proximal tissue responses to radiations commonly found in space. We will achieve this by quantitatively measuring the appropriate variables that represent these processes and using the experimental data to construct a mathematical model of the relationships between the measured variables. The constructed model will provide us with a mechanistic understanding of the effect of radiation quality on tissue responses. Experiments are specifically designed to obtain information on how track structure modifies the radiation induced signaling pathways important to maintaining tissue homeostasis. Using a 3D skin tissue model, we will determine changes in gene expression and protein activation levels for targeted matrix signaling components as a function of radiation quality for two constant beam fluences. Ion beams to be used are: 600 MeV/u iron, 400 MeV/u silicon and 250 Mev/u oxygen. We will use immunohistochemistry to monitor overall matrix integrity as well as cell proliferation and apoptosis rates. We will use the quantitative data collected to determine the parameters of a coarse-grained model involving the key modulators of matrix remodeling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will provide fundamental mechanistic information on the responses of skin to radiation.
Task Progress:	The goal of our NASA project, NNX10AB06G, was to use a combined experimental and computational approach to develop a predictive understanding of the link between cell signaling, gene expression, and proximal tissue responses to radiations commonly encountered in space. We hypothesized that radiation quality differentially affects the tissue response by altering signaling pathways and cellular responses. We characterized molecular, cellular, and tissue level changes induced by radiation – an essential step to construct an integrated understanding of the response. This enabled us to identify the critical biological endpoints for further investigation. We observed radiation quality-dependent effects on endpoints related to the cancer development processes including proliferation, differentiation, and extracellular matrix (ECM) remodeling. We also constructed a predictive agent based model to understand regulation of proliferation and differentiation following high LET exposures. All experiments were performed with a commercially available three dimensional (3D) skin equivalent composed of normal human epidermal keratinocytes and dermal fibroblasts. This is a robust experimental model for mechanistic studies of radiation effects on epithelial tissues. To study radiation quality dependent effects, we exposed the skin model to low linear energy transfer (LET) radiation X-rays, and a range of high atomic number ions (HZE). All ion exposures were conducted with constant ion fluences (3.6 x 10(4) and 1.1 x 10(5) particles/cm2). By following this irradiation protocol, we can focus on the delta ray contribution to the induced tissue responses. We initially studied the global temporal response of the skin model to low LET X-rays using microarray-based transcriptional profiling. This data provided a baseline for comparing high LET responses, and revealed the features and biological processes relevant for radiation responses. We identified cell-type specific responses occurring in the epidermal and dermal layers. The data suggested that ionizing radiation affects cell proliferation and the differentiation profile in the tissue, and affects ECM remodeling. Our data demonstrates that HZE radiation induces LET, and fluence dependent changes in the fraction of proliferating cells, marked changes in the thickness of the epidermal layer, and altered cell proliferation and/or differentiation in the tissue. The largest effect on epidermal thickness was for Neon ion exposures. We analyzed the results using a mathematical model that links cell behavior changes to the experimentally measured tissue-level quantities. Analysis with the agent-based model suggested that radiation-induced changes in both cell proliferation and differentiation are required to match the experimental observations. We have subsequently validated these findings by measuring mRNA expression levels for proliferation and differentiation markers. To correlate observed changes in differentiation with skin function, we measured barrier function following X-ray and Fe ion exposures. Neither radiation disrupted the skin barrier in the first 72 h following exposure. All high LET data endpoints were compared to low and medium doses of low LET X-ray radiation (3 to 200 cGy). We discuss in particular the role of differentiation, emphasizing how the disregulation of cellular signaling in normal tissue may impact progression towards radiation induced health issues in astronauts. Data on mRNA expression, signaling and metalloprotease activities viewed in the context of the ECM remodeling pathway indicated that connective tissue growth factor (CTGF) and plasmin were important drivers of ECM changes in the system. Following HZE ion exposures, we observed differential effects on CTGF intracellular localization in the epidermal and dermal layers as a function of LET. We used a qRT-PCR array containing 84 genes in the ECM remodeling pathway to perform targeted studies of the radiation quality dependence of these biological endpoints, and found strong LET-dependence in the response. While, the ECM pathway was found to be affected both at high and low LET, there was little overlap in the specific transcriptional targets. Interestingly, high LET exposures exhibited a very different cell type specific response than X-rays where dermal responses predominated. HZE exposures resulted in almost a complete absence of transcriptional changes in the dermis. This is likely due to the inhomogeneity of the energy deposition pattern for high LET radiation, and the low density of cells (fibroblasts) in the dermal layer (in our skin model the cell density is ~85% lower in the dermis compared to the epidermis). A manuscript is currently in preparation regarding this work. We made significant progress towards our objective of understanding the tissue response to radiations commonly encountered in space. This grant contributed to five publications, one web article, and two additional manuscripts in preparation. Publications associated with this project: C. von Neubeck, H. Shankaran, N. J. Karin, P. M. Kauer, W. B. Chrisler, X. Wang, R. J. Robinson, K. M. Waters, S. C. Tilton, and M. B. Sowa, Cell type dependent gene transcription profile in a three dimensional human skin tissue model exposed to low doses of ionizing radiation: Implications for medical exposures, Environ. Mol. Mut. 53 (2012) 247-59. Nominated for best paper by young investigator.

	H. Shankaran, T. J. Weber, C. von Neubeck and M. B. Sowa, Using imaging methods to interrogate radiation induced cell signaling. Rad. Res. 177 (2012) 496-507. K. M. Waters, D.L. Stenoien, M.B. Sowa, C. von Neubeck, W.B. Chrisler, R. Tan, R.L. Sontag, and T.J. Weber, Annexin A2 modulates radiation-sensitive transcriptional programming and cell fate. Rad. Res. 179, (2013) 53-61. C. von Neubeck, H. Shankaran, M.J. Geniza, P.M. Kauer, R.J. Robinson, W.B. Chrisler, and M.B. Sowa, Integrated Experimental and Computational Approach to Understand the Effects of Heavy Ion Radiation on Skin Homeostasis, Integrative Biology 5 (2013) 1229-1243. Selected as “hot topic” and for back cover of journal. Tilton, S.C., Weber, T.J., Webb-Robertson, B.J.M., Shankaran, H., Sowa, M.B., Stenoien, D.L., Morgan, W. F., and Waters, K.M. (2014). Data Integration Reveals Key Homeostatic Mechanisms Following Low Dose Radiation Exposure. Toxicology and Applied Pharmacology, (Submitted) Pesantez P, CH Freiin Von Neubeck, WB Chrisler, MB Sowa, and JH Miller. Modeling Selective Irradiation of Epidermal Layers in a Dynamic Skin-tissue Model, 2014. Int. J. Rad. Bio. (submitted) Pirkkanen J, von Neubeck C, Sowa, M. Summary of an integrated experimental and computational approach to study the effects of heavy ion exposures on skin. 06/25/14. https:// (accessed 5/16/17)
Bibliography Type:	Description: (Last Updated: 09/23/2019)
Articles in Peer-reviewed Journals	Waters KM, Stenoien DL, Sowa MB, von Neubeck C, Chrisler WB, Tan R, Sontag RL, Weber TJ. "Annexin A2 modulates radiation-sensitive transcriptional programming and cell fate.mming and cell fate" Radiat Res. 2013 Jan;179(1):53-61. Epub 2012 Nov 13. http://dx.doi.org/10.1667/RR3056.1 ; PubMed PMID: 23148505 , Jan-2013
Articles in Peer-reviewed Journals	von Neubeck C, Shankaran H, Geniza MJ, Kauer PM, Robinson RJ, Chrisler WB, Sowa MB. "Integrated experimental and computational approach to understand the effects of heavy ion radiation on skin homeostasis." Integrative Biology. 2013 Oct;5(10):1229-43. http://dx.doi.org/10.1039/c3ib40071a ; PubMed PMID: 23925568 , Oct-2013
Articles in Peer-reviewed Journals	Pirkkanen J, von Neubeck C, Sowa M. "Summary of an integrated experimental and computational approach to study the effects of heavy ion exposures on skin." The Health Risks of Extraterrestrial Environments (THREE). June 25, 2014. https://three.jsc.nasa.gov/articles/062514_Pirkkanen_et_al.pdf , Jun-2014
Articles in Peer-reviewed Journals	Tilton SC, Matzke MM, Sowa MB, Stenoien DL, Weber TJ, Morgan WF, Waters KM. "Data integration reveals key homeostatic mechanisms following low dose radiation exposure." Toxicol Appl Pharmacol. 2015 May 15;285(1):1-11. Epub 2015 Feb 2. http://dx.doi.org/10.1016/j.taap.2015.01.019 ; PubMed PMID: 25655199 , May-2015
Articles in Peer-reviewed Journals	von Neubeck C, Geniza MJ, Kauer PM, Robinson RJ, Chrisler WB, Sowa MB. "The effect of low dose ionizing radiation on homeostasis and functional integrity in an organotypic human skin model." Mutat Res. 2015 May;775:10-8. http://dx.doi.org/10.1016/j.mrfmmm.2015.03.003 ; PubMed PMID: 25839759 , May-2015